

A Reproducible Benchmark of Relational Database Access-Layer Performance in Express.js: A Configuration-Specific Comparison on PostgreSQL and MySQL

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Abstract

Context: Node.js/Express services reach relational databases through a spectrum of access layers (native drivers, query builders, and object-relational mappers, or ORMs), yet practitioners choose between them largely on vendor benchmarks that are fragmented, PostgreSQL-only, and rarely report the tail (p99). **Objective:** We present a reproducible, byte-equivalence-checked benchmark harness and use it to compare eleven access-layer implementations under one clearly specified operating point on PostgreSQL and MySQL; the harness and its controls, not a generalized library ranking, are the contribution. **Methods:** An identical Express application is served through nine idiomatic access layers (native drivers `pg` and `mysql2`, the query builder Knex, and the ORMs Drizzle, Prisma, Sequelize, TypeORM, Objection and MikroORM) and two tuned native baselines, over five access patterns, on the latest stable library versions. We report throughput with client-observed response-time p99 from 25 repeated runs per cell under three operating conditions: equal demand (a saturated 50-connection point), equal utilization (an open-loop sweep), and an equal compute budget; an automated cross-check verifies byte-identical responses before measurement, and inserts run under the engines' de-

fault durability. **Results:** The implementation-and-strategy difference concentrates in relation-materializing patterns: the deep/nested fetch spreads $7.15\times$ (PostgreSQL) and $4.96\times$ (MySQL) between the native driver and the slowest ORM, narrowing to $1.68\times$ and $2.03\times$ once every layer runs identical SQL: a compound control showing the idiomatic eager-loading strategy drives most of it, isolating no single factor. Read rankings transfer across engines (Spearman $\rho \geq 0.86$); aggregation and writes transfer only moderately ($\rho = 0.64, 0.68$). No layer exceeds $\approx 110\%$ of one core: the native driver leads essentially every pattern and the Rust-free Prisma 7 client sits mid-pack. An earlier version found Prisma tying native at five cores: an artifact of a now-superseded generation, retired by one re-run. Large saturated-p99 gaps are a capacity effect that matched utilization largely closes. **Conclusion:** Access-layer rankings are perishable, as the retired five-core result shows; the durable contributions are the reproducible, byte-equivalence-checked harness, its three-estimand design, and a checklist of benchmarking pitfalls specific to this genre, released as a replication package.

Keywords: database access layer; object-relational mapping; Node.js; reproducible benchmarking; response-time tail (p99); empirical software engineering

1 Introduction

Express.js remains the most widely used web framework for Node.js, and almost every non-trivial service built on it eventually reaches a relational database through some *access layer*. The choice is not settled by the language or the framework: access layers trade raw performance for ergonomics, compile-time type safety, and protection from pitfalls such as the N+1 query problem, where a naive nested fetch silently issues one query per parent row instead of one for the whole result graph [9], a pervasive anti-pattern in real-world database-backed applications (Section 2). Native drivers expose the wire protocol directly (`pg` for PostgreSQL, `mysql2` for MySQL); query builders assemble SQL programmatically (Knex); and object-relational mappers (ORMs) map rows to objects and manage relationships automatically, following patterns such as Data Mapper and Active Record [21]. Node.js back-end performance under load is itself a recognized empirical topic [6], [7], [34], but that literature treats the runtime and web framework as the unit of comparison rather than varying the database access layer.

In practice, the most visible performance comparisons are *vendor benchmarks*: plentiful but narrow, published by the maintainers of the compared products, confined almost exclusively to PostgreSQL, and settling on a single metric family rather than reporting throughput and the tail together. In the sources searched, we did not identify one reporting the p99 tail behavior that is widely held to matter for user-perceived latency under load [19], [22]. The peer-reviewed literature is sparser still, limited to at most three ORMs, confined to PostgreSQL, and, where it benchmarks PostgreSQL against MySQL directly, holding the access layer fixed rather than varying it [2], [15], [48], [62], [63] (Section 2 surveys each).

In the sources we searched (Section 2) this leaves a gap along four axes: (1) no access-layer comparison we found includes MySQL, and no PostgreSQL/MySQL benchmark varies the access layer; (2) no vendor-independent study spans the taxonomy broadly, since the broad-coverage comparisons are vendor-authored and academic work covers at most three libraries [2], [63]; (3) we did not identify one reporting throughput and the p99 tail jointly, the sole vendor suite that pairs throughput with a percentile stopping at a single P95 at one load point [20]; and (4) none of the located studies measures client-observed performance through an HTTP framework, academic studies instrumenting query execution inside the application process [63] and the Express-based work that measures the round trip comparing runtime optimizations on a single stack rather than access layers [28].

This paper addresses that gap with a single, vendor-independent benchmark: we serve an *identical* Express application through all nine access layers (two native drivers, Knex, and six ORMs: Drizzle, Prisma, Sequelize, TypeORM, Objection, MikroORM) over five CRUD-spanning access patterns rather than a measured production workload (point read, keyset range scan, deep/nested fetch, per-author aggregation, and insert; Section 3). Seven of the nine layers (all but the two engine-specific native drivers) run against *both* engines under an identical schema, dataset, and pool size; two tuned native-driver baselines (prepared-statement variants of `pg` and `mysql2`) mark what hand-tuned driver use achieves. Every (layer, engine, pattern) combination reports throughput (req/s) and saturated client-observed response-time p99 (at a 50-connection operating point) jointly, from 25 repeated runs, with a utilization-controlled open-loop tail measured below saturation. The study separates two estimands: the *default-configuration* effect of each layer used idiomatically (RQ1–RQ3), and a controlled *same-SQL* contrast that bounds the residual difference once every layer executes identical SQL, changing query formulation, protocol, preparation, round-trips, and mapping together rather than isolating any one. The full harness (adapters, seed data, an autocannon-

driven [16] load runner, and an automated cross-check that all layers return *byte-identical* responses on the four non-mutating patterns) is released as an open replication package.¹

Three research questions structure the study, each scoped to the benchmarked implementations under the specified operating point:

- RQ1** (*performance difference*): How large is the throughput and response-time-p99 *difference* of the benchmarked implementations relative to the native-driver baseline, how does it vary across the five access patterns, and how does it depend on the operating condition: equal external demand, equal utilization, or an equal compute budget?
- RQ2** (*engine dependence*): Does the observed layer ordering transfer across PostgreSQL and MySQL, or is it engine-dependent?
- RQ3** (*consequential patterns*): For which access patterns is the choice of implementation most consequential in this configuration?

This paper makes four contributions. The **primary** contributions are the harness and the benchmark design; the third is a configuration-specific snapshot of a fast-moving ecosystem, and the harness and design outlast the pinned versions:

- An **open, reproducible harness**: a uniform adapter contract, a deterministic seed, a matrix runner with repeated randomized-block runs and paired request streams, a physical write-state reset, a same-SQL control with server-side protocol evidence, a coordinated-omission-corrected open-loop generator with a utilization-controlled latency mode, and an automated **correctness cross-check** asserting **byte-identical** responses across layers on the four non-mutating patterns (Section 3).
- A **vendor-independent, broad-coverage benchmark design** spanning the taxonomy’s three tiers (native driver → query builder → six ORMs), **dual-engine** (PostgreSQL and MySQL), measured at the **Express level**, reporting throughput and response-time p99 **jointly**, the combination missing from prior art.
- **Configuration-specific observations** on the current library versions and this host: that no layer is multi-core-bound and the native driver leads

¹Harness, deterministic seed, all adapters, raw per-cell measurements, and the table/figure-generating scripts: <https://github.com/mmiothk/express-db-access-performance>; release v1.4.1 is permanently archived at Zenodo, <https://doi.org/10.5281/zenodo.21373621>.

on essentially every pattern, the idiomatic implementation-and-strategy difference concentrating in relation-materializing patterns; that read rankings transfer across engines while aggregation and writes transfer only moderately (visible only in a dual-engine design); and, as direct evidence that such rankings are **perishable**, that a headline finding of an earlier version of this study — the query-engine ORM Prisma tying the native driver at roughly five cores of CPU — **did not survive** re-running the same harness against Prisma’s Rust-free successor, which draws about one core and sits mid-pack (Section 4).

- A practitioner-oriented **checklist of four pitfalls specific to access-layer benchmarking** (Section 5), surfaced during development, several caught by the correctness cross-check and the rest by performance analysis.

2 Related Work

We organize prior work along four axes on which our study differs from it (taxonomy coverage, engine coverage, whether throughput and tail latency are reported jointly, and vendor independence), summarized in Table 1. We located this work with a structured scoping search (not a systematic review) of the ACM Digital Library, IEEE Xplore, Scopus, and Google Scholar, run in July 2026 over titles, abstracts, and keywords. The query crossed an access-layer set (*ORM*, “object-relational mapping”, “query builder”, “database access layer”, and the seven library names) with a context set (*Node.js*, *Express*, *JavaScript*, *TypeScript*), an engine set (*PostgreSQL*, *MySQL*, “relational database”), and a measurement set (*benchmark*, *performance*, *throughput*, *latency*), restricted to 2006–July 2026 and followed by backward and forward citation chaining from the closest hits. We retained empirical comparisons of relational access layers or engines and excluded non-empirical, non-relational, and single-library studies. The exact query strings, per-source search dates, and screening decisions are archived with the replication package (`notes/related-work-search.md`); because this is a scoping rather than an exhaustive search, we scope every resulting gap claim to the sources searched.

2.1 Object-relational mapping and the access-layer spectrum

Access layers exist to bridge the object-relational impedance mismatch between the relational model and the object graphs an application manipu-

lates [29], [46], spanning native drivers, query builders, and object-relational mappers following patterns such as Data Mapper and Active Record [21] (Section 1). Torres et al. [54] survey two decades of ORM patterns; the trade-off they document, developer productivity against a runtime cost that is hard to predict, is the one this paper quantifies. The same question recurs in the polyglot-persistence debate over non-relational stores [47]; we hold that choice fixed and vary only the access layer.

2.2 ORM performance and data-access anti-patterns

A parallel line of research studies ORM performance from the opposite direction: rather than benchmarking layers, it detects and repairs the inefficient SQL that mapping frameworks emit. The costliest pattern is the N+1 query problem (Section 1); it and related redundant-query anti-patterns are pervasive across real-world web applications [58], [59], and Shao et al. [50] give an independent taxonomy and detector for database-access anti-patterns. Chen et al. quantify the performance impact of redundant data access in ORM-backed Java systems [10], extending earlier anti-pattern detection work [9]. A complementary strand shows the overhead is often *removable*: query synthesis lifts imperative accessor code into efficient SQL [12], lazy evaluation batches round trips [11], view-centric optimization rewrites request handlers [60], and second-level caching absorbs repeated reads [8]; the JavaScript analogue is mis-sequenced `async/await` serializing independent I/O [55]. Closest to our design, Lorenz et al. [41] show the best O/R mapping strategy depends on the database technology, van Zyl et al. [65] that ORM performance is highly sensitive to tuning, and a recent study adds an energy dimension [5]. This literature establishes *where* ORM overhead concentrates (relation-materializing, N+1-prone access) and that it is configurable rather than fixed, which our per-pattern results confirm; it does not compare drivers, query builders, and ORMs head-to-head under controlled load, our aim here.

2.3 Peer-reviewed access-layer comparisons

The peer-reviewed literature that measures access layers head-to-head is narrow. An early study by van Zyl et al. [64] weighed object databases against ORM tools. Colley et al. [15] give the earliest systematic cross-language account, benchmarking eight ORM frameworks across Java, C#, Python, and PHP against a common schema, but *deliberately excludes JavaScript*, leaving Node.js, and Express in particular, without an academic baseline for the better part of a decade. Three recent studies partially address this: one, in *Procedia Computer Science*, compares Prisma against optimized raw SQL

over eight query patterns on a containerized PostgreSQL setup [62] (cited for its methodology; differing library versions, hardware, workload, and harness mean we neither build on nor dispute its absolute speedup figures); a second compares Prisma, TypeORM, and Sequelize across four relation types, also on PostgreSQL, by response time, memory, and CPU [2]; and the most recent instruments the same three ORMs on PostgreSQL inside a NestJS service, timing internal query stages rather than client-observed requests [63], independently observing the concurrency-dependent Prisma ranking flip we also report (worst at single queries, best under parallel load). A further Express-based study optimizes a single stack (pooling, batching, caching) rather than comparing access layers [28].

2.4 Database-engine, SQL/NoSQL, and Node.js performance

A second, largely disjoint line of work benchmarks the layers immediately below and above the access layer without varying it directly. Salunke and Ouda [48] compare PostgreSQL and MySQL head-to-head under standard OLTP workloads (the only dual-engine precedent found), but benchmark the *engines themselves*, bypassing any application-level access layer, so their work cannot say how a driver, query builder, or ORM would change it. In the broader engine-comparison line, work contrasting SQL and NoSQL databases [40] measures the store, not the access layer. On the Node.js side, Chaniotis et al. [7] established Node’s viability as a web platform, Kuffel and Walter [34] tracked how its back-end performance drifts under sustained load, do Carmo et al. [6] compared back-end frameworks more broadly, and Sun et al. [52] characterize how the single-threaded event loop schedules the I/O every access layer relies on. None isolates the database access layer as an independent variable, let alone varies it across the taxonomy studied here.

2.5 Standard benchmarks and benchmarking methodology

Standard benchmarks define how database systems are measured: YCSB [18] for cloud-serving stores and the TPC-C/TPC-E suites for OLTP. These target the *engine* and its transaction mix, saying nothing about the driver, query builder, or ORM through which an application issues work (the variable this study isolates), but they set the methodological bar for controlled, repeatable load. A separate methodological literature explains why the vendor benchmarks’ single-metric habit is a problem in its own right: Fruth et al. [22]

and Raasveldt et al. [45] catalog pitfalls specific to database benchmarking (coordinated omission, warm-up, tool-induced distortion) that silently corrupt reported percentiles; Dean and Barroso [19] show why the tail, not the mean, determines user-perceived latency once a request fans out over many components, as an Express request serving a deep/nested fetch does; and Li et al. [39] trace the hardware, OS, and application sources of that tail. Load-testing surveys report such measurement is often ad hoc [30], [38], and repeatability studies find systems results frequently hard to reproduce [14], motivating a shared, reusable harness; the benchmarking literature stresses relevance, repeatability, and fairness [27] and the role of neutral benchmarks in advancing a field [51]. From this literature we take a design choice absent from every access-layer benchmark surveyed above: report throughput *and* tail latency (p99) for the same run, rather than a single metric family chosen after the fact.

2.6 Practitioner and vendor benchmarks

A larger body of evidence comes from practitioners and vendors, though each source is authored by a party with a stake in one of the compared products. Drizzle publishes both a Northwind micro-suite spanning nine targets (pg, Knex, Kysely, MikroORM, TypeORM, Prisma, and Drizzle itself, pooled and unpooled) and an official k6-driven load test reporting requests per second, average latency, P95, and CPU [20]. Prisma’s own site compares Prisma, Drizzle, and TypeORM by median query latency over 500 iterations, reporting neither throughput nor any latency percentile [43]. IMD-Bench, maintained by Gel Data (formerly EdgeDB), adds Sequelize and node-postgres and reports both throughput and latency on three CRUD-style operations [23]. Together these three sources cover more of the taxonomy than the entire peer-reviewed literature combined, yet share the same defects noted in Section 1: vendor authorship, PostgreSQL-only coverage, and a single reported metric family.

2.7 Positioning

In the sources searched through July 2026 (search protocol archived with the replication package), no study combined broad taxonomy coverage, both engines, joint throughput/tail reporting, and vendor independence; each covers at most two or three of these four properties (Table 1). This study combines all four (Section 1), closing the same engine gap separating access-layer studies from engine-only work such as [48].

Table 1: Prior work on relational-database access-layer performance, positioned along the four axes on which this study differs from it. “Layers covered” aggregates native drivers, query builders, and ORMs into one count; “none” marks studies that vary the engine or framework but not the access layer.

Study	Layers covered	Engines	Metrics	Independent?
Colley et al. [15]	8 ORMs, 4 languages (no JS)	not stated	query latency	Yes
Procedia CS [62] [†]	Prisma vs. raw SQL	PostgreSQL	latency, CPU	Yes
JCCT [2]	3 ORMs	PostgreSQL	latency, memory, CPU	Yes
JCSI [63]	3 ORMs	PostgreSQL	per-stage latency	Yes
Salunke and Ouda [48]	none (engine only)	PostgreSQL + MySQL	throughput, latency	Yes
Drizzle [20]	9 (broad)	PostgreSQL	throughput, latency, P95	No (vendor)
Prisma [43]	3 ORMs	PostgreSQL	median latency only	No (vendor)
IMDBench [23]	4 ORMs + driver	PostgreSQL	throughput, latency	No (vendor)
This study	9 idiomatic (+2 tuned)	PostgreSQL + MySQL	throughput + response-time p99	Yes

[†]Cited for its methodology only; its absolute speedup figures are not comparable to ours given the differing versions, hardware, workload, and harness.

3 Study Design

Our goal, in Goal–Question–Metric terms, is to *analyze* the relational database access layer of an Express.js service *for the purpose of* quantifying its performance cost *with respect to* throughput and response-time p99 *from the point of view of* an application developer choosing a layer *in the context of* PostgreSQL and MySQL back ends [4]. The design follows established guidelines for controlled experimentation in software engineering [32], [57], crossing seven portable access layers with both engines and all five patterns, plus two engine-specific native drivers and their tuned counterparts `pg-tuned` and `mysql2-tuned` (eleven layers: nine idiomatic, two tuned reference baselines); the full rationale is distributed with the replication package (`METHODOLOGY.md`). Two estimands are kept separate throughout: the *default-configuration* estimand (RQ1–RQ3, Section 4), the cost a practitioner incurs from each layer used idiomatically, and a *same-SQL (raw-path)* estimand, what remains once every layer executes identical SQL through its raw facility, a compound contrast rather than a mechanism-isolating decomposition (Section 4).

A fixed concurrency imposes equal *demand* on layers of unequal capacity and therefore drives them to unequal *utilization*, so the performance-difference question (RQ1) is characterized under three operating conditions rather than one privileged point. *Equal external demand* is the saturated closed-loop matrix, where every layer receives the identical request stream at a fixed 50-connection load (Section 4). *Equal utilization* is the coordinated-omission-corrected open-loop sweep, where each layer is offered a fixed fraction (50/70/85/95%) of its *own* saturating throughput. An *equal compute*

budget is the equal-CPU control, where the server is confined to a fixed number of cores. These conditions answer different questions and need not agree: the saturated condition measures capacity under overload, the matched-utilization condition the near-intrinsic per-request latency, and the equal-budget condition the compute cost of the same work. Treating any one as *the* operating point would conflate these, so we report all three and say which each result speaks to.

3.1 Factors and treatments

We vary three factors. The *access layer* has eleven levels spanning the taxonomy: the native drivers `pg` and `mysql2`, their tuned counterparts `pg-tuned` (named prepared statements reused across calls) and `mysql2-tuned` (the binary protocol with server-side prepared statements via `execute()`), included as engine-specific tuned reference baselines rather than idiomatic treatments; the query builder Knex; and the ORMs Drizzle (a lightweight, query-builder-style ORM), Prisma, Sequelize, TypeORM, Objection, and MikroORM. We classify a library by its dominant interface, not its self-description (Section 1); Drizzle sits at the query-builder end of the ORM range and Objection layers an ORM over Knex, which the results bear out. The nine idiomatic layers cover the taxonomy’s three tiers but are a representative cross-section, not an exhaustive catalogue; other libraries (for example Kysely, Slonik, and pg-promise) are omitted. Vendor independence here means authorship only: no evaluated library’s maintainers were involved or invited to inspect the adapters, which does not remove the consequential choices any benchmark author makes (library selection, idiomatic-use criteria, schema and endpoints, pool size, allowed tuning); we disclose these throughout and release the harness for inspection. The *engine* has two levels, PostgreSQL 18.4 and MySQL 9.7.1, and the *access pattern* has five levels, realized as HTTP endpoints (Table 2). The four native and tuned baselines are engine-specific (`pg/pg-tuned` run only on PostgreSQL, `mysql2/mysql2-tuned` only on MySQL); the other seven layers run on both engines, giving $4 + 7 \times 2 = 18$ layer $\times$ engine cells and $18 \times 5 = 90$ measured configurations. Supplement Table S27 summarizes which factors are held constant, which vary, and which are uncontrolled on this single virtualized host; the uncontrolled factors are why we report results for this configuration rather than as general library performance.

Table 2: The five access patterns and what each stresses.

Endpoint	Pattern	Stresses
GET /posts/:id	point read (PK)	per-query overhead
GET /posts?before=	keyset range scan	index seek + hydration
GET /posts/:id/thread	deep/nested fetch	join strategy, N+1 avoidance
GET /authors/:id/summary	aggregation	grouped counts / raw SQL
POST /posts	insert	write path

3.2 Objects: schema, workload, and data

The workload is a minimal blog domain (`authors 1-* posts 1-* comments *-1 authors`), exercising a one-to-many relationship and a two-hop join for the deep fetch. The five endpoints implement canonical CRUD access patterns [18] (Table 2): the keyset page issues `WHERE id < cursor ORDER BY id DESC LIMIT n` against the primary-key index rather than a large `OFFSET`; the deep/nested fetch returns a post with its author and every comment, each with its own comment-author; and the aggregation reports post count, comment count, and total views per author. The database is loaded from a deterministic seed (2,000 authors, 100,000 posts, 1,000,000 comments) by the native driver, independently of any layer under test, so every engine receives the same seed *data payload* (byte-identical row values); the on-disk physical representation, index layout, and collation naturally differ between the two engines. The working set fits in memory, so the measurements emphasize the access layer’s per-request instruction, protocol, and mapping cost with disk-read latency largely removed; engine-side query execution, buffer management, locking, and logging remain part of every measurement [26]. Every request draws its target identifier uniformly at random from the full seeded range (posts 1–100,000, authors 1–2,000), so load spreads across the whole working set rather than repeatedly hitting one hot record, and the read endpoints measure warm access-layer cost. A skewed, production-like key distribution is a planned variant (Section 6).

3.3 The adapter contract and N+1 control

Every access layer implements the same factory interface (five query methods plus a teardown), so the Express server and the load runner are layer-agnostic. Each adapter returns an *identical result shape*, and each uses that layer’s *recommended idiomatic* mechanism to avoid the N+1 query problem on the deep fetch: a hand-written join for the native drivers, the Knex query builder, and

the query-builder-style Drizzle (the tuned baselines reuse the native drivers’ join unchanged), and relation eager loading (`include`, `withGraphFetched`, `populate`, or `relations`) for the data-mapper ORMs. The comparison is therefore one of *idiomatic usage*, not a fixed query plan: a measured difference reflects the SQL each layer generates, its round-trip count, and how it materializes results. One asymmetry is inherent to any driver-versus-ORM comparison: the native side’s idiom is hand-authored SQL, whereas the ORM side’s idiom is the library’s default. Server-side statement logging captures the exact SQL each layer emits, its round-trip count, and `EXPLAIN` plans for the native statements, released with the replication package; Supplement Table S2 tabulates the per-endpoint counts, which differ by design: the deep fetch takes one query in Sequelize and MikroORM, two in the native drivers, Knex, Drizzle, and TypeORM, and four in Prisma and Objection. The idiomatic API for each layer was fixed by a rule decided *before* measurement—the eager-loading API its official documentation presents first for the pinned version—so the treatment is a documented choice, not an editorial one; Supplement Table S16 and the package’s `METHODOLOGY.md` record, per layer, that API with its **documentation page and version**, its round-trip count, the documented alternative we did *not* use, and that query logging, hooks, and validation are off. The six ORMs split both ways on the join-versus-select-in default; a loading-strategy sensitivity check (Section 4) switches join-default data-mapper ORMs to select-in wherever the alternative is a byte-identical drop-in. To separate a layer’s execution machinery from its strategy choice, every adapter also implements a *same-SQL control*: identical two-statement deep-fetch SQL (and, per engine, identical `EXPLAIN` plans) with identical row mapping, executed through the layer’s raw-SQL facility, mirroring the aggregation control (Section 4). Server-side statement capture, synchronized to a request marker so connection handshakes are excluded, confirms the two control statements are byte-identical across every layer on both engines; what differs is the wire protocol each driver negotiates for them: on PostgreSQL every layer uses the extended protocol with server-side prepared statements except MikroORM, which falls back to the simple protocol with client-side literal inlining, and on MySQL every layer uses the text protocol except `mysql2-tuned` and Prisma, which use the binary protocol with server-side prepared statements. To ensure no adapter silently returns less, or issues $N+1$ queries, an automated cross-check (`bench/verify.mjs`) funnels every adapter’s rows through shared canonical constructors and asserts full JSON byte equality against the native-driver baseline, on both engines, across 12 probes per adapter spanning the four non-mutating patterns and the same-SQL control, before any timing is collected. This check caught two defects during development (a fan-out aggregation bug and a driver result-shape

mismatch; see Section 5).

3.4 Controls

Four controls isolate the access layer as the only variable. (i) *Pool size* is fixed at ten connections for every adapter, so the comparison measures access-layer overhead rather than pool tuning, a known web-application performance factor [25], [36]; for Prisma, whose default pool is otherwise derived from core count, this is pinned explicitly via the `connection_limit` URL parameter. Because the load generator opens 50 concurrent connections against this ten-connection pool, roughly forty in-flight requests are always waiting in each library’s connection-acquisition queue; this queueing is deliberately part of what we measure and is held identical (same pool size, demand) across layers. A 200 ms connection-pool sampler corroborated this on the deep fetch: the native PostgreSQL cell ran with its ten-connection pool fully occupied (mean utilization ≈ 10 of 10) and a mean of about 32 requests pending in the acquisition queue (peaks of 39–40). (ii) *Identical query semantics*: each endpoint issues the same logical query across layers; aggregation uses a single correlated-subquery formulation everywhere (the documented raw-SQL path), so it measures layer overhead on an identical plan, not differences in generated SQL. (iii) *Write isolation*: because the insert endpoint mutates the dataset, every cell resets it before warm-up and before every measured run, not merely once per cell; the write endpoint is additionally measured in its own dedicated server boot, taken after the database’s physical state is rebuilt rather than merely emptied by row deletion (PostgreSQL: drop and recreate the database from a seeded template; MySQL: delete the benchmark rows, rebuild the table with `OPTIMIZE TABLE`, and re-pin `AUTO_INCREMENT`), so sequences, physical layout, and purge state cannot drift across replicates or leak between adapters. (iv) *Observer separation*: the HTTP load generator runs in a process separate from the server.

3.5 Measurement procedure

Load was generated with `autocannon` [16], an HTTP/1.1 load tester issuing requests continuously at fixed concurrency and recording a latency histogram; it is a closed-loop (constant-concurrency) generator [49]. We measured each of the 90 configurations as 25 *repeated runs*. For each replicate we booted a fresh server process, opened 50 concurrent connections, ran a fifteen-second warm-up pass whose measurements were discarded, and then took one twelve-second measured run. The warm-up length follows a separate cold-boot measurement for a fast, middling, and the

slowest PostgreSQL layer: every layer reached and sustained at least 95% of steady-state throughput within 12 s (Prisma by 5 s, native by 9 s, MikroORM by 12 s), absorbing JIT compilation, pool fill, and plan caching, since managed runtimes do not always reach a stable steady state promptly [3]. The experimental unit is the freshly booted server run for one (layer, engine, endpoint) cell in one replicate; individual HTTP requests within a run are subsamples, not independent units. Booting a fresh process per replicate prevents persistent application-process state (JIT state, pool contents, heap) from carrying across replicates, though not full statistical independence: all runs share one host and one short campaign, so replicate index and measurement order are treated as blocking factors, not sources of independence. Cell order was reshuffled independently within each replicate; a forward-versus-reversed-order check on the write endpoint (5 replicates per direction) found no detectable history effect (reversed-to-forward throughput ratio across all 18 cells: 0.970–1.021, median 1.004). Within a replicate, every layer and engine is driven by the identical request-id sequence, generated by a PRNG seeded from the (endpoint, replicate) pair; this paired-stream design removes between-layer sampling noise as a confound, and the first ids of every stream are archived in the replication package. We report the median of the 25 replicate values; 449 of the 450 dedicated write-endpoint boots in the primary campaign completed normally, the exception being knex/PostgreSQL, whose replicate-16 boot hit a health-check timeout immediately after the write-state reset, leaving that cell with 24 of 25 repeated runs (the runner records and skips such failures rather than retrying silently). Each replicate yields throughput (requests per second) and latency percentiles p50, p90, p97.5, and p99. During measured runs we sampled server CPU and peak resident memory from `/proc` over the full process tree (parent plus descendants, so an in-process engine or spawned helper cannot escape measurement), with database and load-generator CPU sampled separately; a `perf_hooks` GC observer (event count, total pause time) and the 200 ms connection-pool sampler ran inside the server, polled through a `/stats` endpoint after each run. Inserts were measured under each engine’s *default* durability configuration (PostgreSQL `fsync=on`, `synchronous_commit=on`, `full_page_writes=on`; MySQL `innodb_flush_log_at_trx_commit=1`, `sync_binlog=1`, `log_bin=ON`), the regime a deployment runs unless an operator deliberately relaxes it. Each insert is an autocommit single-statement `INSERT` under each engine’s default isolation level (PostgreSQL Read Committed, MySQL Repeatable Read); the transactional variant wraps a post and its five comments in one transaction. A labelled secondary run instead relaxed every one of those settings on both engines (10 replicates, write endpoint only) to

isolate the durability mechanism, compared with the default regime in Section 5 (Supplement Table S3). A fixed-payload `/baseline` endpoint returning a fixed, seed-realistic thread-shaped payload measured the shared Express-plus-serialization floor (bounding framework and serialization cost, not per-request object construction): 10,909 (PostgreSQL server) and 10,921 (MySQL server) req/s, roughly $3\times$ the fastest deep-fetch cell, so the framework floor leaves between-layer spreads visible rather than compressed. To characterize behavior under load, we swept the deep/nested fetch (the most layer-sensitive pattern) across connection counts from 1 to 256 for every layer and engine, since throughput and contention scale non-linearly with concurrency [61]; a separate equal-CPU-budget control (1, 2, or 4 cores) attributes part of that scaling directly to CPU (Section 4). A native, `undici`-based constant-arrival generator drove the deep fetch under a fixed offered rate (250–4,000 req/s) rather than fixed concurrency, measuring latency from each request’s intended send time and clipping timed-out requests into the distribution at their cutoff; this coordinated-omission-corrected design [33], [53] independently validates the closed-loop tail results (Section 4).

3.6 Analysis

We separate a small set of *primary* outcomes, which carry the paired inference and the research-question claims, from *secondary* and *exploratory* outcomes, which are reported descriptively as controls and sensitivity analyses (Table 3). The primary outcomes are throughput and the closed-loop p99 tail on the five patterns against both engines at the fixed operating point; everything else bounds or contextualizes them.

Because absolute throughput is hardware-specific, we analyze *relative* differences across layers within an engine. We quantify a pattern’s sensitivity to the access layer by its *spread*: the idiomatic native driver’s (`pg` or `mysql2`; the tuned baselines are a separate tuned reference, not the spread’s anchor) median throughput divided by that of the slowest layer on that pattern (native-relative, so comparable across patterns even where an ORM exceeds the native driver). We report medians over the 25 repeated runs, the coefficient of variation (CV) as a stability signal, following recommended practice [24] and using enough replicates to bound variance [35] (Section 4), and a bootstrap 95% CI for the median. The design is a randomized block: within each replicate every layer runs the identical request stream, so two layers’ throughput samples are matched by replicate and must be compared with *paired* tests. To test whether adjacently ranked layers are distinguishable rather than an artifact of measurement noise, we compare each adjacent

Table 3: Outcomes and their analysis role. The primary outcomes carry the paired inference and the research-question claims; the secondary and exploratory outcomes are reported descriptively as controls and sensitivity analyses that bound or contextualize the primary results, and are not treated as additional confirmatory hypotheses.

Role	Outcome	Analysis
Primary	Throughput (req/s), 5 patterns $\times$ 2 engines, idiomatic layers, 50-connection operating point	Paired permutation, Wilcoxon, bootstrap median CI; spread & rank
Primary	Closed-loop p99 tail latency, same cells	Paired permutation, Wilcoxon, per-cell bootstrap CI on p99
Secondary	Same-SQL (raw-path) contrast	Bounds the residual (compound) same-SQL difference
Secondary	Sub-saturation open-loop tail (coordinated-omission corrected)	Near-intrinsic latency companion to the saturated primary p99
Secondary	Layer $\times$ engine interaction (per pattern)	Blocked permutation (F on D)
Secondary	Combined app+db CPU efficiency, equal-CPU control	End-to-end compute view; operational confirmation of the CPU trade
Explor.	Aggregation fan-out; OFFSET vs. keyset; pool-size sweep; durability & isolation sensitivity; RSS	Descriptive; pitfall illustration and robustness checks

pair on the deep/nested fetch on their per-replicate throughput ratios with a two-sided paired sign-flip permutation test (20,000 permutations, so the smallest attainable p is $1/(20,001)$), cross-checked with the Wilcoxon signed-rank test. We report the geometric-mean paired ratio with a paired bootstrap 95% CI and the paired dominance (fraction of replicates in which the faster layer wins). In every test the unit is the *run-level* statistic (one throughput or p99 value per repeated run, 25 per cell), never the individual request; all confidence intervals are *percentile* bootstraps resampling replicate indices (preserving the pairing), from a fixed seed (`mulberry32(0x57a75)`) so every resample is reproducible. Because p99 is reported at millisecond resolution and contains ties, the permutation test keeps zero log-ratios and the Wilcoxon test drops them with the standard tie correction. Because the seven adjacent pairs are defined by the *observed* ranking rather than preregistered, we treat their significances descriptively (post-hoc adjacent-rank comparisons), applying a Bonferroni correction as a conservatism check rather than claiming a confirmatory family. For the pg-Prisma pair we additionally report a paired two-one-sided-tests (TOST) equivalence result against a $\pm 5\%$ margin on the log-ratio (justified below). The layer $\times$ engine interaction is tested per pattern with a blocked permutation test: for each layer and replicate we form the paired PostgreSQL-minus-MySQL log-throughput difference $D_{\ell,i}$ and, under the null of no interaction, permute the layer labels of $\{D_{\ell,i}\}$ *within* each replicate i (20,000 permutations); the test statistic is the between-layer F of a randomized-block ANOVA on D with replicate as the block. Rank agreement

across engines (Spearman, Kendall) is reported descriptively over the seven evaluated layers — systems under study, not a sample from a population — so we do not attach population confidence intervals to those correlations. Reporting nonparametric tests with standardized effect sizes follows recommended practice for randomized performance experiments in software engineering [1], [13]. The $\pm 5\%$ equivalence margin is an author-selected smallest effect of practical interest for *this* study, not a general engineering threshold: at modest deployment scale a throughput difference under 5% would not change a capacity-planning decision (for example, how many application instances to provision for a target load), and it is within the run-to-run and day-to-day variation typical of a virtualized host (this study’s own run-to-run CV reaches 7.6%); at very large scale even 5% can correspond to many machines, so the margin should be reset per deployment. We chose it before the analysis but did not preregister it, and the equivalence conclusion is not sensitive to the exact value within a few percent.

3.7 Experimental setup

All measurements were taken on 2026-07-16 and 2026-07-17 on a single host: the primary matrix ran overnight, and the order-invariance, durability, same-SQL, open-loop, fan-out, equal-CPU, concurrency-sweep, utilization, pool-size, cluster, mixed-workload, wait-event, and post-restart runs followed (a virtualized Intel Xeon Gold 6140 instance, 16 vCPUs, single NUMA node, 126 GB RAM; Linux 6.17; Node.js 24.18.0; Express 5), with the database engines (PostgreSQL 18.4, MySQL 9.7.1) running locally. Each library is pinned to the *latest stable release compatible with the harness adapter contract* as of the freeze date (2026-07-15), recorded from the committed lockfile and an automated environment capture (Supplement Table S11); a library is pinned to an earlier release only where the latest stable breaks the contract or the byte-equivalence cross-check, and any such deviation is documented per layer. Only Sequelize invokes this exception: its version-7 line is a pre-release, so the current 6.x stable line is used. Prereleases and unreleased point versions are otherwise excluded. Because access-layer performance is version-sensitive, this pins a dated snapshot rather than a permanent ranking, exactly one headline of which a superseded pin produced and a re-freeze retired (Section 4).

3.8 Replication package

The harness — the Express application, the adapter contract, all nine idiomatic adapters and the two tuned baselines, the deterministic seed, the

`autocannon` matrix runner, the correctness cross-check, and the scripts re-generating every table in Section 4 — is released so the full matrix can be re-run with one command against a containerized or local PostgreSQL and MySQL.

4 Results

The tables report throughput (requests per second, higher is better) and tail latency (p99, milliseconds, lower is better) per access layer and engine. The two consequential patterns are in the main text (the deep/nested fetch, Table 4, and the insert, Table 5); the point read, keyset range scan, and aggregation are in Supplement Tables S12, S13, and S15. All numbers come from the run of Section 3. Each native driver is reported both idiomatically (`pg`, `mysql2`) and tuned (`pg-tuned`, `mysql2-tuned`), the tuned rows marking the throughput ceiling hand-tuned driver use reaches, not a level every idiomatic layer attains without code changes. The comparison of interest is *relative* across layers within an engine (Section 3); two estimands recur below: the *default-configuration (idiomatic) effect* (RQ1–RQ3) and the *same-SQL effect* (Supplement Table S14).

Table 4: Deep/nested fetch: throughput (req/s, higher is better; median with a within-campaign 95% bootstrap interval over the 25 repeated runs — run-to-run variability within one host and campaign, not across hardware or days) and response-time p99 (ms, lower is better) by access layer and engine.

Layer	Category	PostgreSQL		MySQL	
		req/s [95% CI]	p99	req/s [95% CI]	p99
<code>pg</code>	native-driver	3687 [3602–3737]	20	–	–
<code>mysql2</code>	native-driver	–	–	2382 [2357–2451]	28
<code>pg-tuned</code>	native-tuned	3746 [3684–3828]	18	–	–
<code>mysql2-tuned</code>	native-tuned	–	–	2415 [2387–2427]	25
<code>knex</code>	query-builder	2487 [2465–2505]	27	2007 [1989–2031]	30
<code>drizzle</code>	orm-lightweight	1865 [1837–1878]	35	1318 [1300–1336]	47
<code>prisma</code>	orm	1186 [1160–1203]	51	634 [618–644]	93
<code>sequelize</code>	orm	991 [977–998]	60	886 [868–898]	67
<code>typeorm</code>	orm	1204 [1188–1223]	50	1017 [987–1029]	57
<code>objection</code>	orm	1028 [1020–1037]	57	834 [828–855]	69
<code>mikroorm</code>	orm	516 [499–525]	116	480 [474–488]	120

Table 5: Insert: throughput (req/s, higher is better; median with a within-campaign 95% bootstrap interval over the 25 repeated runs — run-to-run variability within one host and campaign, not across hardware or days) and response-time p99 (ms, lower is better) by access layer and engine.

Layer	Category	PostgreSQL		MySQL	
		req/s [95% CI]	p99	req/s [95% CI]	p99
pg	native-driver	6402 [6242–6547]	11	–	–
mysql2	native-driver	–	–	1753 [1717–1763]	113
pg-tuned	native-tuned	6413 [6073–6571]	11	–	–
mysql2-tuned	native-tuned	–	–	1750 [1482–1759]	120
knex	query-builder	4841 [4629–4908]	15	1670 [1470–1692]	127
drizzle	orm-lightweight	4085 [3928–4146]	15	1730 [1324–1745]	120
prisma	orm	2774 [2750–2815]	23	886 [828–903]	153
sequelize	orm	2732 [2719–2747]	23	1478 [1375–1684]	125
typeorm	orm	2648 [2630–2694]	23	1345 [1301–1359]	125
objection	orm	3907 [3746–3980]	17	1728 [1598–1746]	117
mikroorm	orm	1979 [1949–1997]	30	1257 [1240–1268]	117

4.1 RQ1: performance difference relative to the native baseline

The native driver leads on essentially every pattern and engine. On PostgreSQL, `pg` (or its tuned counterpart) is fastest on all five patterns; on MySQL, `mysql2` is fastest on the two reads, the deep fetch, aggregation, and the insert. The current-generation ORMs sit mid-to-low throughout: Prisma, in particular, is seventh of the eight layers on both point reads, fifth on the PostgreSQL deep fetch and seventh on the MySQL deep fetch, lowest-ranked on aggregation (last on MySQL, seventh on PostgreSQL, above only `Objection`), and last on the MySQL insert (Tables 4, 5; Supplement Tables S12, S13, and S15). An earlier generation of this benchmark reported Prisma tying or leading the native driver at a large CPU premium; on the current stable versions that finding does not hold, and the resource subsection below shows why: no layer now exceeds about one application core, so there is no CPU-for-throughput trade to buy parity with.

`pg-tuned` and `mysql2-tuned` track their idiomatic counterparts within a few percent on the single-round-trip patterns and edge them only on the deep fetch, where issuing more than one round trip lets reusable prepared statements and MySQL’s binary protocol pay off (`pg-tuned` 3,746 vs. `pg` 3,687; `mysql2-tuned` 2,415 vs. `mysql2` 2,382); on PostgreSQL aggregation `pg-tuned` adds about 8% (7,538 vs. 6,978).

On the **point read**, the native-relative spread is $2.78\times$ on PostgreSQL (pg 6,835 down to MikroORM 2,455) and $2.11\times$ on MySQL (mysql2 4,368 down to MikroORM 2,067; Supplement Table S12); the native driver leads on both engines, and Prisma is next-to-last (3,255 / 2,074), ahead of only MikroORM.

On the **deep/nested fetch** (the pattern that assembles a post, its author, and all comments with their authors), the native-relative spread is the widest measured: $7.15\times$ on PostgreSQL (pg 3,687, 95% CI [3,602–3,737], down to MikroORM’s 516 [499–525]) and $4.96\times$ on MySQL (mysql2 2,382 [2,357–2,451] down to MikroORM’s 480 [474–488]). The native driver leads outright on both engines — there is no tie at the top — with Knex (2,487 / 2,007) and the lightweight ORM Drizzle (1,865 / 1,318) next, Prisma mid-pack on PostgreSQL (1,186, fifth) but near the bottom on MySQL (634, seventh), and the remaining data-mapper ORMs, especially MikroORM, trailing furthest. The comparison is paired (Section 3); every adjacent step of the MySQL ranking separates under the paired permutation test (Table 7, discussed below).

Aggregation is the least layer-sensitive pattern: the native-relative spread is only $1.83\times$ on PostgreSQL and $1.72\times$ on MySQL (Supplement Table S15), and the native driver leads both. The idiomatic difference is small when a layer forwards a single query and large only when it must materialize a nested object graph application-side, as on the deep fetch. Prisma is the slowest layer on the MySQL aggregation (2,328) and second-to-last on PostgreSQL (4,408, above only Objection’s 3,812).

4.2 Same-SQL control

The idiomatic deep fetch lets each layer choose its own query strategy, mixing raw SQL execution with everything else a layer does. The *same-SQL control* (Section 3) bounds this residual difference: server-side capture confirms the statement text is byte-identical across layers on both engines, while the wire protocols differ per layer; this control is therefore *same SQL, protocol disclosed*, not *same plan*. The contrast is composite: switching from a layer’s idiomatic relational-fetch path to the common raw-SQL-and-shared-mapping path changes its query strategy, query count, prepared-statement behavior, the raw versus ORM API, and result marshalling together, so the residual difference is bounded but no single factor is isolated.

On the identical SQL, the deep-fetch native-relative spread collapses from its idiomatic $7.15\times$ (PostgreSQL) and $4.96\times$ (MySQL) to $1.68\times$ and $2.01\times$ (Supplement Table S14). Most of the idiomatic spread is therefore the eager-loading *strategy*, not the execution machinery: on the raw path Drizzle rises from 1,865 to 3,237 req/s, Prisma from 1,186 to 2,155, and Knex from 2,487

to 2,926 on PostgreSQL, so layers that trail on the idiomatic fetch converge once they run the same statement. Prisma is the slowest layer on the raw path itself (2,155 on PostgreSQL, 1,206 on MySQL), which we attribute to dispatch and marshalling overhead in its raw API rather than SQL execution. MikroORM shows the largest strategy gap: its idiomatic path reaches only 0.16 (PostgreSQL) and 0.20 (MySQL) of its own same-SQL throughput, a $6.3\times$ and $5.1\times$ idiomatic-to-raw gap. A loading-strategy sensitivity check (Supplement Table S18) tests whether these deficits are merely a default-strategy artifact: switching the join-default data-mapper ORMs (Sequelize, MikroORM) to select-in, where the alternative is a byte-identical drop-in, makes them slower ($0.73\text{--}0.80\times$), so their deficit is intrinsic; conversely, Objection’s batched-select-in default is about $1.6\times$ slower than a single join on MySQL (820 vs. 1,301 req/s), so part of *its* deficit there is a strategy choice rather than an intrinsic cost.

4.3 RQ2: engine dependence of the ranking

The relative ordering of access layers is largely, but not uniformly, stable across engines. The Spearman rank correlation between the PostgreSQL and MySQL orderings of the seven portable layers (descriptive; Section 3) is high on the reads — $\rho = 0.86$ on the point read, 0.89 on the range scan, and 0.86 on the deep fetch — so a layer that reads well on one engine reads well on the other. It is only *moderate* on aggregation ($\rho = 0.64$) and on the insert ($\rho = 0.68$): these two patterns transfer across engines less reliably, though neither is uncorrelated. Prisma is the notable mover: mid-pack on PostgreSQL inserts (2,774 req/s, fourth of the seven portable layers) but last on MySQL inserts (886, behind even MikroORM’s 1,257) — a rank flip invisible to a single-engine benchmark (Supplement Table S26). The *magnitude* of a layer’s cost also depends on the engine: a blocked permutation test on the paired PostgreSQL-minus-MySQL log-throughput difference (Section 3) finds the layer $\times$ engine interaction significant on all five patterns (observed statistic above every one of the 20,000 permutations). Because the cross-engine differences are large and systematic, that p reaches its floor ($1/20,001$) and conveys little about magnitude; the magnitude is largest on the insert, where Prisma’s throughput falls from 2,774 req/s on PostgreSQL to 886 on MySQL. Single-engine results are thus portable in order for reads but not in degree, most loosely for writes.

On the **keyset range scan** the native driver again leads, with a native-relative spread of $4.21\times$ on PostgreSQL and $3.85\times$ on MySQL (Supplement Table S13); the large nominal spread is driven disproportionately by MikroORM alone (905 / 856 req/s, far below every other layer), confirming

that the earlier collapse observed under `OFFSET` pagination was a workload artifact rather than an engine or layer property (Section 5).

Inserts are the pattern where the engine dominates most visibly, and where the durability regime matters to that claim: all headline write numbers here are under each engine’s *default* durability configuration (Section 3). On MySQL the faster layers cluster tightly near an engine-imposed floor — `mysql2` 1,753, `Drizzle` 1,730, `Objection` 1,728, and `Knex` 1,670 req/s, within about 5% (Table 5) — with p99 around 113–153 ms for every layer; `Prisma` is the exception at 886 req/s (last), which widens the nominal MySQL spread to 1.98×. Direct `performance_schema` instrumentation attributes the floor to the commit path: the redo-log and binlog flush is a shared engine cost, similar across the native driver, query builder, and slowest ORM despite their very different read performance, and removing it (relaxed durability) lifts `mysql2` from 1,576 to 3,238 req/s (Supplement Table S19). Relaxing durability roughly doubles MySQL insert throughput layer-wide (`mysql2` 2.35×, 1,753 to 4,114; Supplement Table S3), confirming the floor is engine-side rather than layer-specific. PostgreSQL, by contrast, spreads 3.23× (`pg` 6,402 down to `MikroORM` 1,979) and is only weakly durability-sensitive on the lean layers (`pg` 1.13× relaxed÷default; Supplement Table S3), so there the access layer, not the engine, remains the larger factor.

4.4 RQ3: which patterns matter most

Ranking the five patterns by native-relative spread on PostgreSQL gives deep fetch (7.15×) > range scan (4.21×) > insert (3.23×) > point read (2.78×) > aggregation (1.83×). On MySQL the head and tail of that order are unchanged — deep fetch (4.96×) > range scan (3.85×) at the top, aggregation (1.72×) last — while the insert and point read swap in the middle (point read 2.11×, insert 1.98×), the insert compressing because MySQL’s engine floor, not the access layer, sets its ceiling (RQ2 above). The deep/nested fetch is thus the most consequential pattern on both engines and aggregation the least; the range scan’s large nominal spread is driven disproportionately by `MikroORM` alone.

The primary deep-fetch workload uses posts with roughly ten comments on average. A fan-out sweep across 0, 1, 10, 50, 100, and 500 comments (both engines, medians of three repeated runs) shows the native-relative spread is *roughly constant* with graph size rather than growing with it, so on this schema the deep-fetch deficit behaves as a fixed multiplier. (An earlier three-run sweep had suggested a growing spread; that did not survive replication, so we report the fan-out as exploratory.)

4.5 Tail latency

We report the closed-loop p99 for every cell (p50, p90, p97.5 in the replication package), measured at the 50-connection operating point against the fixed ten-connection pool (Section 3), so the tail includes connection-acquisition queueing — the tail a deployment sees when driven to saturation, not an estimate of intrinsic per-request latency; the open-loop experiment below reports the sub-saturation tail. Under saturation, tail latency tracks throughput: on the deep/nested fetch, PostgreSQL’s p99 ladder runs **pg** 20, Knex 27, Drizzle 35, TypeORM 50, Prisma 51, Objection 57, Sequelize 60, and MikroORM 116 ms (Table 4), monotonic below the native driver — a penalty visible in the saturated tail that throughput-only vendor benchmarks miss. The tail ordering mirrors the throughput ranking, which separates pairwise on both engines (Table 7). On MySQL inserts, by contrast, the faster layers’ tails cluster (113–127 ms) at the engine floor, with only Prisma’s 153 ms standing out — the tail counterpart of the throughput engine floor.

These closed-loop values come from a constant-concurrency generator; to check the tail ranking is not an artifact of that model, we drove the deep/nested fetch on PostgreSQL under the coordinated-omission-corrected open-loop harness (Section 3; timed-out requests, a 10 s hard cap, clipped into the distribution; full sweep in Supplement Table S6). Below saturation the corrected tails are flat and small: **pg** holds p99 at 2–3 ms up to 2,000 req/s, and Knex and Prisma hold single- to low-double-digit tails up to 1,000 req/s (Knex 2 ms, Prisma 13 ms). Beyond each layer’s capacity the tail collapses rather than degrading gracefully: at 2,000 req/s Prisma and Sequelize exceed 10 s while **pg** still clears requests at 3 ms and Knex at 89 ms, and the data-mapper ORMs collapse earliest—Sequelize at 2,000 req/s and MikroORM already at 1,000. At the 4,000 req/s tier **pg**’s corrected p99 is 1.48 s while Knex reaches 8.8 s and the ORMs exceed 10 s. We call a layer’s behavior at an offered rate a *collapse* when it achieves under half the offered rate with a large share of requests timing out. By that criterion the ordering matches the closed-loop result: the data-mapper ORMs enter unstable queueing as soon as offered load crosses their capacity, while the lightweight layers degrade gradually. The same open-loop sweep on MySQL (Supplement Table S17) reproduces the pattern—flat sub-saturation tails and capacity-crossing collapse—so the saturated-tail ordering is engine-robust. That saturated tail tracks capacity, not intrinsic latency. Offering each layer 50/70/85/95% of *its own* saturating throughput (coordinated-omission-corrected, five runs, both engines; Supplement Tables S21–S22), the tails converge: at 50% utilization every layer holds p99 near 2–5 ms on both engines, regardless of where it sits in the saturated ladder, and rises only as it nears

its own capacity (for example Objection’s p99 reaches 285 ms at 95% of its own capacity on PostgreSQL). At matched utilization the large saturated gap (pg 20 ms versus MikroORM 116 ms) therefore largely dissolves; it is a capacity-and-queueing effect, not a difference in intrinsic tail latency. The single-rate open-loop cells above are single runs, so near-knee point values are indicative; the utilization sweep and the flat-then-collapse ordering are what reproduce.

4.6 Measurement stability and significance

Across the 25 repeated runs the coefficient of variation of deep-fetch throughput — the pattern that carries the pairwise tests — is at most 3.8% on either engine (Supplement Tables S1 and S9), well below the between-layer spreads reported above; dispersion is larger only on the insert (up to 6.6% on MySQL). Every measured request in the primary matrix succeeded: HTTP errors, connection-acquisition timeouts, and non-2xx responses were zero across all 90 cells’ measurement windows (nonzero timeouts arise only in the deliberate open-loop overload runs below). Those non-2xx counters are load-bearing, not decorative: on the current versions MikroORM 7 removed `persistAndFlush`, so every MikroORM *write* threw an HTTP 500 — which returns faster than a real insert, so the broken cells briefly showed *inflated* throughput and spuriously led the MySQL insert until the runner’s non-2xx counts flagged them. The read-path byte-equivalence check had passed, because the defect was on the write path; correctness must be asserted before timing on *every* path, not only the one an equality check exercises. As an environmental-robustness check we restarted both database engines (cold buffer pools, fresh connections) and re-measured a representative deep-fetch subset (pg/mysql2, Prisma, MikroORM): the layer ranking reproduced the primary campaign on both engines, with absolute throughput up to 16% lower eight days later (larger on PostgreSQL), day-to-day host drift the relative analysis absorbs (Supplement Table S20).

Comparing each adjacently ranked pair on the MySQL deep/nested fetch with the paired permutation test (method in Section 3; Table 7), all seven pairs separate: six with the observed statistic above every one of the 20,000 permutations ($p = 1/20,001 \approx 5 \times 10^{-5}$) and the faster layer ahead in every replicate (dominance 1.00), all with ratios at least 1.15; the narrowest step, Sequelize over Objection, still separates (ratio 1.05, dominance 0.88, $p = 0.0001$). The Wilcoxon test agrees throughout (replication package). Because the pairs are adjacent ranks of the *observed* ordering rather than preregistered, we treat these significances descriptively; a Bonferroni correction leaves all seven far below any conventional threshold. Native mysql2

leads outright with no tie at the top, so every step of the ranking is both statistically and practically supported.

4.7 Scaling with concurrency

A concurrency sweep from 1 to 256 connections (Supplement Figure S1) shows the three-band ordering (native driver, then Knex and Drizzle, then the data-mapper ORMs) emerges by 8 connections and holds at every operating point above about 32; at a single connection the layers are far closer together (about 320–1,020 req/s) and Prisma is mid-pack throughout, never approaching the native driver. The 50-connection operating point used throughout sits above the knee for every layer, so the RQ1 ranking is not an artifact of a single point.

4.8 Resource footprint

Throughput is not the only cost (Table 6). On the current library generation the application-side CPU cost is uniform: on the deep/nested fetch, measured at the same ten-connection pool as every other layer, every layer draws about one core (100–110% of one logical core, the maximum being MikroORM’s 110%), so no layer buys throughput with extra application cores. What differs is the database-side load each layer generates. The native `pg` driver runs at roughly 100% application CPU but issues lean SQL that drives the database harder — about 355% database CPU on the PostgreSQL deep fetch — whereas the ORMs sit at the same $\approx 100\%$ application CPU and drive lower database CPU (on MySQL, `mysql2` draws 85% database CPU against 25–70% for the ORMs; Table 6). The native driver therefore leads on throughput and on end-to-end compute efficiency at once: counting combined application-plus-database CPU per completed request, `mysql2` completes 1,254 req per combined-CPU-second on the MySQL deep fetch against Prisma’s 384 and MikroORM’s 356 (Supplement Table S5). There is no CPU-for-throughput trade: the layers that are slower are also less compute-efficient.

An equal-CPU control confirms that every layer is application-single-core-bound. Confining the server process with `taskset` to 1, 2, or 4 cores (database and load generator pinned to disjoint cores; Supplement Table S4) changes deep-fetch throughput by at most a few percent for every layer: `pg` holds 3,689, 3,688, and 3,687 req/s at 1, 2, and 4 cores, Prisma 1,079, 1,052, and 1,219, and MikroORM 446, 495, and 522. No layer’s per-process throughput improves with extra cores, so the per-layer differences are not an artifact of unequal core budgets. Because each layer is bound within one process, throughput scales out horizontally instead: under a shared-port `node` cluster

Table 6: Resource footprint on the deep/nested fetch (MySQL, medians over 25 replicates): application-process-tree CPU and database-server CPU (% of one logical core), end-to-end throughput per combined (application+database) CPU-second, and peak resident memory. No layer exceeds about 110% of one application core: the native driver runs near 100% application CPU but issues lean SQL that drives the database server harder, while the ORMs sit near 100% application CPU at lower database cost, so there is no CPU-for-throughput trade. Per-layer efficiency and the equal-CPU control are in Supplement Tables S5 and S4.

Layer	app CPU%	db CPU%	req/comb-CPU-s	RSS (MB)
mysql2	105	85	1254	294
knex	100	75	1147	215
drizzle	105	50	850	297
prisma	105	60	384	334
sequelize	100	30	682	262
typeorm	100	70	598	225
objection	100	45	575	221
mikroorm	110	25	356	356

(Supplement Table S23) Prisma rises from 1,117 to 2,275 to 4,449 req/s at 1, 2, and 4 workers on PostgreSQL, reaching native-like aggregate throughput at four workers (pg 3,299 to 4,055 to 3,961, saturating near two workers as it becomes database-bound). Prisma’s low single-process throughput is thus recoverable by running more processes, at roughly four times the cores.

Memory differences are smaller (Table 6): the query builder and the lightest ORMs are cheapest (Knex 215, Objection 221, and TypeORM 225 MB peak RSS), while Drizzle, Prisma, and MikroORM use more (297–356 MB). Per-instance resource use governs how many replicas a given load requires when sizing horizontally scaled deployments [56].

5 Discussion

Access-layer rankings are perishable; the instrument is the contribution. The strongest lesson of this study is one its own method forced on us. An earlier campaign, run on the library generation current at the time, found the ORM Prisma (then version 5.22, built on an in-process Rust query engine) tying the native driver on the deep/nested fetch while drawing about five application cores — a striking result we reported as a configuration-

Table 7: Paired comparison of adjacently ranked access layers on the deep/nested fetch (MySQL), over the 25 repeated runs. Because every layer runs the identical per-replicate request stream, layers are compared on their per-replicate throughput ratios: median throughput (req/s) with bootstrap 95% CI, the geometric-mean paired ratio A/B with a paired bootstrap 95% CI, paired dominance (fraction of replicates in which A exceeds B), and the two-sided paired sign-flip permutation p (20000 permutations; a value of 5.0×10^{-5} is the resolution floor $1/(B+1) = 1/20,001$, i.e. the observed statistic exceeded every permutation; the Wilcoxon signed-rank test agrees, replication package).

Comparison (A > B)	med. A [95% CI]	med. B [95% CI]	ratio A/B [95% CI]	dom.	perm. p
mysql2 > knex	2382 [2357–2451]	2007 [1989–2031]	1.19 [1.18–1.20]	1.00	5.0×10^{-5}
knex > drizzle	2007 [1989–2031]	1318 [1305–1336]	1.53 [1.51–1.55]	1.00	5.0×10^{-5}
drizzle > typeorm	1318 [1300–1336]	1017 [987–1029]	1.30 [1.28–1.32]	1.00	5.0×10^{-5}
typeorm > sequelize	1017 [987–1029]	886 [868–898]	1.15 [1.13–1.16]	1.00	5.0×10^{-5}
sequelize > objection	886 [868–898]	834 [828–855]	1.05 [1.03–1.07]	0.88	0.0001
objection > prisma	834 [828–855]	634 [618–641]	1.33 [1.31–1.36]	1.00	5.0×10^{-5}
prisma > mikroorm	634 [618–644]	480 [476–488]	1.31 [1.29–1.33]	1.00	5.0×10^{-5}

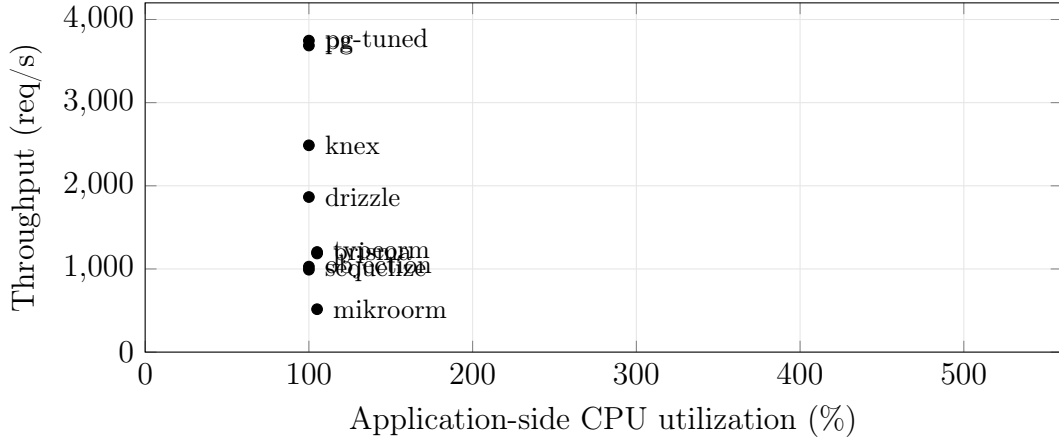


Figure 1: Throughput versus application-side CPU on the deep/nested fetch (PostgreSQL, ten-connection pool, medians of 25 replicates). Every layer sits at about 100–110% of one application core; the native driver leads on the throughput axis, and the slower ORMs are low on throughput at the same CPU. CPU is the server process only; database-side CPU is excluded for all layers alike.

specific observation. Re-running the *identical* harness, workload, and host against the latest stable releases (Prisma 7.8, now a Rust-free TypeScript client executing through a JavaScript driver adapter) retired that finding outright. No layer now draws more than about one application core (the maximum application-side CPU across all 90 cells is 110%), and Prisma has fallen to mid-to-low throughput across the matrix: seventh of eight layers on both reads on each engine, fifth on the PostgreSQL deep fetch and seventh on MySQL, near the bottom of the aggregation ranking on both engines, and last on MySQL inserts (Section 4). Nothing changed but one dependency’s internals across a single major version. This is the paper’s central claim made concrete: access-layer performance is highly sensitive to implementation and version [65], so any specific ranking is perishable, and the durable contributions are the reusable harness, the benchmark design, and the pit-falls checklist that let a current ranking be re-established as the ecosystem moves rather than read from a frozen table.

Implications for practice. RQ1 and RQ3 agree that abstraction cost is specific to the access *pattern*, concentrating where a layer must hydrate rows into objects and resolve associations rather than merely issue SQL (the object-relational impedance mismatch [29] made concrete). The native-relative spread is the widest measured on the deep/nested fetch — $7.15\times$ on PostgreSQL and $4.96\times$ on MySQL — and narrows to $1.83\times$ and $1.72\times$ on aggregation, where every layer forwards the same correlated-subquery plan (Section 4). The same-SQL control bounds this cost rather than localizing it to any one mechanism (Section 4). Round-trip count alone does not set throughput: on the PostgreSQL deep fetch the single-query MikroORM is the slowest layer of all (516 req/s), while the two-query Drizzle (1,865) outruns both the two-query TypeORM (1,204) and the four-query Prisma (1,186) (Supplement Table S2; Table 4), so how a layer materializes a result graph matters more than how many statements it issues.

The same-SQL control. Switching each layer from its idiomatic relation-fetch path to a common raw-SQL path sharply compresses the deep-fetch field: on PostgreSQL Drizzle rises from 1,865 to 3,237 req/s, Knex from 2,487 to 2,926, and Prisma from 1,186 to 2,155 (Section 4), so most of the idiomatic spread reflects each layer’s eager-loading strategy rather than its raw execution machinery. Because the contrast changes query formulation, round-trip count, prepared-statement behavior, wire protocol, and result marshalling together, it bounds the residual difference without isolating a single factor. Server-side capture confirms the statement text is byte-identical across layers

while the wire protocols differ per layer (Section 3), so protocol choice alone does not account for the collapse.

Compute cost and horizontal scaling. Under current versions the throughput differences are no longer bought with extra application cores. On the deep fetch every layer sits at roughly one application core — Prisma at about 105%, the native driver at about 100% (Figure 1) — and the native driver’s lead comes instead from issuing lean SQL that pushes work onto the database tier (its PostgreSQL database-side CPU reaches about 355% against Prisma’s 60%). The equal-CPU control confirms that each layer is essentially single-core-bound per process: from one to four cores the native driver holds 3,689, 3,688, 3,687 req/s and Prisma 1,079, 1,052, 1,219, so neither recovers much throughput from extra cores within one process (Section 4). A layer’s low per-process throughput is therefore not a fixed ceiling: because Prisma is single-core-bound per worker, it scales out nearly linearly under a shared-port node cluster, rising from 1,117 to 2,275 to 4,449 req/s at one, two, and four workers on the PostgreSQL deep fetch and reaching native-like aggregate throughput at four workers (Section 4) — at four times the processes and cores, trading footprint and energy for throughput [5].

Queueing under load and the open-loop check. The open-loop cross-run (Section 4) shows that a closed-loop benchmark alone would underestimate how abruptly, not just how much, a heavyweight ORM’s tail degrades once offered load crosses its capacity.

Writes: a partly engine-bound cost under default durability. RQ2 has a clear answer on inserts, measured under each engine’s *default* durability configuration (Section 3): the native driver leads on both engines, and PostgreSQL’s insert throughput remains strongly layer-dependent even with fsync on (3.23× from pg at 6,402 down to MikroORM at 1,979 req/s; Table 5). On MySQL the spread is narrower (1.98×): most layers cluster near an engine-level commit floor — the native driver and the fastest ORMs fall within about 5% of one another at 1,670–1,753 req/s — consistent with the per-commit double flush (redo log, then binlog) InnoDB performs on every insert [26], though Prisma sits well below that cluster (886 req/s, last on MySQL inserts). The write ranking transfers only moderately across engines (Spearman $\rho = 0.68$, against 0.86–0.89 for the reads; Section 4). This warrants caution: a write comparison run on a single engine invites the wrong generalization either way (“the access layer barely matters to writes” from

MySQL, or “it matters a great deal” from PostgreSQL), and an engine-only benchmark such as [48] cannot expose this interaction, since it omits the access layer entirely; only the dual-engine design does, consistent with earlier quantitative ORM work showing database technology modulates mapping-strategy cost [41]. Relaxing durability does not overturn the engine contrast (Section 4). The single-row insert is not the only write shape: a transactional multi-statement write, inserting a post and five comments in one transaction, spreads the representative layers about $3.7\times$ (native pg 2,718 down to Prisma 733 req/s; Supplement Table S8), below the widest read spread, because the transaction’s single commit amortizes the per-commit flush across six inserts. Prisma, mid-pack on the single-row PostgreSQL insert, falls to the bottom of the transactional subset, so the transactional ordering does not simply track the single-row one.

Methodological lessons: a checklist for access-layer benchmarking.

Reaching the figures above meant confronting four confounds that would otherwise have corrupted the comparison silently. One, the aggregation fan-out, is a *correctness* bug our automated byte-equality cross-check caught, along with a driver result-shape mismatch and — on the write path, which that read-oriented check does not cover — a fresh, current instance: MikroORM 7 removed `persistAndFlush`, so every MikroORM insert threw a 500, and because an error response returns faster than a real insert, the broken cells briefly appeared to *lead* the MySQL write ranking until the runner’s non-2xx response counter flagged them and excluded the cell. Timing is meaningless until every response is a real 2xx, so the error counter is the write path’s correctness gate as byte-equality is the read path’s. The other three confounds return *correct* results, invisible to any equality check, and were found only by performance analysis. We regard the resulting checklist as reusable:

- **Aggregation fan-out.** A naive join between `posts` and `comments` evaluated before the `SUM` multiplies each post’s view count by its number of matching comment rows, inflating the aggregate, a correctness bug invisible unless every layer’s output is checked against a reference, not a performance one.
- **OFFSET vs. keyset pagination.** An early, OFFSET-based design for the range-scan endpoint collapsed on MySQL at realistic offsets, which would have been misread as an access-layer weakness rather than what it is: a pagination-strategy cost that InnoDB pays more dearly for than PostgreSQL. Rewriting every adapter around keyset pagination (`WHERE id < cursor ORDER BY id DESC LIMIT n`) removed the confound.
- **Write-workload contamination.** The insert endpoint grows `posts` on

every call; because engines and layers commit at different rates, later cells in a run would scan a differently sized table depending on run order rather than on layer identity, unless benchmark-inserted rows are deleted before every cell.

- **Query-formulation confounds.** A first attempt at the aggregation endpoint pre-aggregated *all* of `comments` before filtering to one author, so every request re-scanned the whole table regardless of which author was requested, a query-plan artifact masquerading as an access-layer effect [37], fixed by rewriting the query as a correlated subquery touching only the target author’s rows.

None of these four is schema-specific; each is a generic failure mode of the genre, extending the database-benchmarking pitfalls cataloged by Fruth et al. [22] and Raasveldt et al. [45] to the access-layer-comparison sub-genre. Each is easy to introduce by accident and hard to detect (the correctness ones without an automated cross-check, the performance ones without profiling every cell), and at least one plausibly explains part of the gap between some vendor-reported and independently verified figures in this space.

Practical guidance. The practical synthesis follows from RQ1–RQ3, on the default-configuration estimand a practitioner faces, and applies to the benchmarked implementations in this configuration, workload, and host rather than to access-layer categories in general. The native driver leads throughput on every pattern on both engines, and the idiomatic ORM penalty concentrates on relation-materializing traffic: the deep/nested fetch spreads $7.15\times$ on PostgreSQL and $4.96\times$ on MySQL (Section 4), whereas read-simple patterns (point reads, keyset scans) and aggregation, where a layer merely forwards SQL, spread far less. Where a service’s hot path is dominated by deep or nested fetches, the access layer is consequential, and a thinner path — the native driver or a query builder — is the safer default *for throughput and response-time p99*, since a heavyweight data-mapper ORM concentrates its largest penalty on that traffic. Where the workload is read-simple or write-bound, especially on MySQL, whose engine floor for single-row inserts dominates layer differences under default durability, the throughput cost is small, so a full ORM’s productivity and type safety, not measured here, may reasonably decide the choice. A layer’s low per-process throughput is not on its own disqualifying: Prisma, among the slowest layers here on several patterns, recovers native-like aggregate throughput by scaling out (1,117 to 4,449 req/s from one to four Node.js workers on the PostgreSQL deep fetch; Section 4), at four times the processes, so the choice trades single-process efficiency for replica count. Second-level

caching can absorb part of an ORM’s read cost where the workload tolerates staleness [8].

These are performance findings only: the access-layer decision also turns on developer productivity, type safety, correctness, maintainability, and security, none of which this study measures, and which may outweigh the throughput differences reported here. Even as performance guidance, the recommendations rest on one schema, one hardware configuration, a single-host local-database deployment measured under each engine’s default durability for writes, and the pinned library versions — which, as the retired Prisma finding above shows, can be superseded within a single major release. The same-SQL comparisons bound the idiomatic cost; they are not a license to replace idiomatic defaults with raw SQL in practice. A production deployment adds network latency, TLS, a reverse proxy, a separate database host, connection poolers, read replicas, and larger or varied payloads, any of which can shift the balance; Section 6 sets out how far these results can generalize.

6 Threats to Validity

We organize the threats along the four standard validity categories [17], [57].

Construct validity. Our dependent variables are HTTP-level throughput (req/s) and tail latency (p99) measured at the Express endpoint, not query-execution time at the driver or database boundary — deliberate, since it captures the full request round trip a deployed service incurs, including Express routing and JSON serialization, but it means the overhead we attribute to the *access layer* may partly reflect these surrounding costs. We control for this by holding those costs near-constant: every adapter satisfies a documented *adapter contract* and funnels its rows through shared canonical constructors (a fixed field set, key order, and value typing, under a fixed TZ=UTC), and an automated cross-check (`bench/verify.mjs`) asserted full JSON byte equality against the native-driver baseline, on both engines, across the four non-mutating patterns before any timing run (Section 3). During development that check caught a defect: PostgreSQL’s schema declared `created_at` as a timezone-naive timestamp although the column is `timestampz`, so Drizzle’s PostgreSQL responses carried instants shifted by the host’s UTC offset — invisible to throughput and to any coarser equivalence check; it was fixed (`withTimezone: true`) before the campaign reported here. The fixed-payload baseline floor sits well above the fastest measured cell (Section 3), so the framework floor compresses, rather than manufactures, between-layer differences, which are therefore attributable to how each layer reaches an

identical response, not to what is returned. Writes are the one endpoint that mutates state; the reported finding uses each engine’s *default*, vendor-standard durability configuration (Section 3), not one we relaxed ourselves, so the write spreads (Table 5) describe a standard deployment rather than a tuning choice; a symmetric relaxed-durability secondary confirms the cross-engine gap is not an artifact of either configuration (Section 5, Supplement Table S3).

Internal validity. A naively configured ORM can trigger $N+1$ queries and appear artificially slow for reasons unrelated to its inherent cost; every adapter must use its layer’s documented, idiomatic eager-loading strategy on the deep-fetch endpoint (Section 3). The correctness cross-check (now the byte-level comparison above) caught two defects during development, both fixed before the measurements reported here (the checklist in Section 5). Explicit warm-up phases per cell (Section 3), set to exceed the slowest layer’s cold-start stabilization, absorb JIT, pool-fill, and plan-cache effects, guarding against environmentally induced measurement bias [42]; two further controls rule out non-layer confounds: benchmark rows reset before every cell (physically rebuilt for writes), and one correlated-subquery plan for aggregation across layers (Section 3). Every layer and engine within a replicate is driven by the identical, PRNG-seeded id sequence, and a forward-versus-reversed cell-order check found no detectable history effect (Section 3) — addressing the concern that processing order could confound layer identity with run position. A residual risk is coordinated omission in the load generator, which can under-report tail latency at saturation [22], [53]; a native, coordinated-omission-corrected constant-arrival cross-run on the deep fetch (Section 3) confirms the tail ranking holds under that stricter model (Section 4; Supplement Table S6).

External validity. All measurements come from a single schema, a single dataset size, one hardware configuration, and a connection pool fixed at ten, against specific engine and library versions (PostgreSQL 18.4, MySQL 9.7.1; Section 3). The fixed pool makes this an *equal configured connection limit* comparison, not a per-layer-tuned one: it isolates the access layer from pool tuning but does not show each layer’s throughput under its own best pool. A per-layer pool-size frontier on both engines (Supplement Tables S7 and S24) addresses this directly: the ordering is preserved from a pool of 1 to 50, and each layer’s throughput saturates well below 50, so the fixed-pool choice does not drive the ranking. A mixed read/write workload (Supplement Table S25) likewise preserves the read ordering. Results may not transfer to

larger working sets, other pool sizes, or other engine versions, and library performance ages quickly, so rather than freeze an arbitrary snapshot the study pins the latest stable release of each library compatible with the harness at the freeze date (2026-07-15; Supplement Table S11). Prisma is therefore measured at 7.8, the Rust-free pure-TypeScript client that replaced the earlier in-process Rust query engine [44], so the results characterize the current architecture rather than a superseded one; an earlier campaign on the Rust-engine generation had reported a multi-core application-CPU profile for Prisma that this re-run retires (Section 5), a concrete demonstration that access-layer rankings can age within a single major release. Two version choices are disclosed rather than resolved: Sequelize is measured on its stable 6.x line because the version 7 line is a prerelease at the freeze date, and Prisma’s MySQL path runs through the MariaDB driver adapter (@prisma/adapters-mariadb), since Prisma ships no mysql2 adapter — an asymmetry with the other MySQL layers. Pinning does not remove the currency risk, because libraries keep moving; what re-establishes a current ranking is the instrument, not the table, as this re-run itself shows. The host is a virtualized, 16-vCPU, single-NUMA-node cloud instance rather than dedicated bare metal, so absolute req/s figures reflect that substrate specifically; only the *relative* ordering within an engine is intended to travel. The working set is memory-resident, so these results are a hot-cache regime maximizing the visibility of layer differences; as a workload becomes I/O-bound (larger than RAM, cold caches), every layer increasingly waits on the same storage and the spreads should compress toward $1\times$. The deep-fetch object graph is small (a post with on average ten comments, at most 25 in the seed); the replicated fan-out sweep (Section 4) shows the spread is a roughly fixed multiplier from 0 to 500 child rows, so the headline figure is representative of the range rather than specific to one graph size. We mitigate these limitations by pinning every version, publishing the seed and an automated environment capture, and releasing the harness for re-runs on other hardware or against newer releases. A further limitation is that the load generator and server share a host, common practice in this genre that removes network variance as a confound but can understate network-bound effects; process separation on one host is not hardware isolation, since the primary campaign did not pin application, database, and generator to disjoint cores, so scheduler and shared-cache contention between them is possible (the environment capture records governor, NUMA, and affinity). A resource-isolation check re-ran the representative deep-fetch cells with the three components pinned to disjoint core sets (ISO_ run, replication package): the isolated and shared-host medians agree within about 2% (ratios 0.98–1.01 across pg, Prisma, and MikroORM), and the layer ordering is unchanged, so same-host

contention is not driving the reported ranking. The equal-CPU control (Section 4, Supplement Table S4) now shows every layer is essentially single-core-bound per process — no layer’s deep-fetch throughput changes materially from one to four cores — so the ranking is not an artifact of unequal core access. A single-process caveat still extends to deployment topology: the server runs as a single Node.js process, whereas production deployments typically run one worker per core (`cluster`, `pm2`, per-pod replicas). A multi-worker check (Supplement Table S23) shows the low-per-process layers scale out near-linearly, so a layer’s single-process throughput is not its aggregate ceiling — Prisma rises from 1,117 to 4,449 req/s from one to four workers on the PostgreSQL deep fetch, reaching native-like aggregate throughput at four times the processes. A sustained-load check bounds temporal drift: holding three representative layers (native `pg`, Prisma, MikroORM) under ten minutes of continuous deep-fetch load, re-run under the final harness, moves throughput by at most $\pm 1.4\%$ between the first and last three minutes, with the band ordering preserved in every single minute, and a 90 s run reproduces the 12 s medians within 1% (replication package), so the short measured runs are not sampling a transient; longer-horizon drift over hours to days [34] remains untested. A two-machine cross-run over a dedicated link, adding real network latency, and a multi-worker variant are planned to bound this further.

Conclusion validity. Our analysis (Section 3) reports medians with the coefficient of variation and bootstrap 95% confidence intervals as stability signals and, because the design is a randomized block, compares adjacently ranked layers with *paired* tests on their per-replicate throughput ratios (paired sign-flip permutation, cross-checked with the Wilcoxon signed-rank test; Table 7). Two bounds reinforce the ordering (Section 4): the widest spreads dwarf the run-to-run CV — deep-fetch throughput varies by at most about 3.8% across the 25 repeated runs on either engine (Supplement Tables S9 and S1) — and the three-band ordering holds across the concurrency sweep at ≥ 32 connections (Supplement Figure S1). On the deep/nested fetch all seven adjacently ranked pairs separate (Table 7): six at the permutation-resolution floor ($p = 1/20,001$) with the faster layer ahead in every replicate, and the closest pair — Sequelize over Objection, a ratio of 1.05 — at $p = 10^{-4}$ with the faster layer ahead in 88% of replicates; all seven survive a Bonferroni correction across the post-hoc adjacent-rank comparisons (Section 4). Rather than assert a specific statistical power, which would require a prospective effect-size and variance model we did not prespecify, we rest the conclusions on effect sizes and interval estimates: the separating pairs combine large paired ratios with the permutation floor

and non-overlapping bootstrap intervals, above the run-to-run dispersion. Booting a fresh process per replicate prevents persistent process state from carrying across replicate runs [31], but the replicates share one host and one campaign; we therefore treat replicate index and measurement order as blocks (Section 3) rather than claiming full independence, and more replicates or longer runs would tighten the intervals further.

7 Conclusion and Future Work

This paper replaces vendor-benchmark claims with a vendor-independent, reproducible comparison of relational database access layers in Express.js, serving an identical application through a representative taxonomy (native drivers, a query builder, and six ORMs) across PostgreSQL and MySQL, over five access patterns, reporting throughput and response-time p99 together (Section 1).

The idiomatic implementation-and-strategy difference is concentrated, not uniform: sharpest for the deep/nested fetch that assembles a nested object graph application-side, and modest for point reads and aggregation (Section 4). The same-SQL control shows this difference is bounded but not attributable to any single mechanism, and that the idiomatic eager-loading strategy drives most of it (Section 4). On the current library versions no layer is multi-core-bound and the native driver leads on essentially every pattern; read rankings transfer across the two engines while aggregation and writes transfer only moderately, a caution against generalizing single-engine results (Section 5). That access-layer rankings are perishable is, here, not a hypothesis but a measured result: a headline of an earlier version of this study, the query-engine ORM Prisma tying the native driver at roughly five cores of CPU, did not survive re-running the same harness against Prisma’s Rust-free successor. Development also surfaced a reusable checklist of benchmarking pitfalls specific to this genre, one of them, a fast-but-wrong write path that passed the read-level cross-check, added by this very re-run (Section 5).

These findings held under a concurrency sweep (1 to 256 connections) and under paired permutation and Wilcoxon tests confirming adjacent-layer separation (Section 4). Further extensions would deepen the evidence: a two-machine cross-run isolating the load generator would bound observer-interference effects beyond the open-loop check performed here; additional engines and dataset sizes would widen external validity; and a dedicated study of the N+1 penalty and of cold-start latency would sharpen the practical guidance. Beyond these findings, the durable contribution is the har-

ness itself: a vendor-independent, dual-engine instrument with a correctness cross-check and a same-SQL contrast control, released under open licenses so that these extensions, and independent re-runs against future library versions (like the one this revision performed against Prisma 7), are straightforward, addressing the repeatability gap widely documented in computer-systems research [14].

Data availability

The complete replication package (benchmark harness, database schema and deterministic seed, all adapter implementations, the byte-level correctness cross-check, raw per-cell measurements, and the scripts that generate every table and figure) is available at <https://github.com/mmio tk/express-db-access-performance> release v1.4.1 is permanently archived at Zenodo (<https://doi.org/10.5281/zenodo.21373621>). This is the generic build; the Elsevier submission build is `ist/ist_main.tex`.

CRedit authorship contribution statement

Mateusz Miotk: Conceptualization, Methodology, Software, Investigation, Data curation, Formal analysis, Visualization, Writing – original draft, Writing – review & editing.

Declaration of competing interest

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Declaration of generative AI and AI-assisted technologies

During the preparation of this work the author used a large language model (Claude, Anthropic) to assist with drafting and editing the manuscript and with implementing and analyzing the benchmark harness. After using this tool, the author reviewed and edited the content as needed and takes full responsibility for the content. All AI-assisted code was verified: the statistical estimators carry unit tests against hand-computed values and known closed-form properties (`bench/stats.test.mjs`), every reported table cell traces to a raw per-cell record through a committed generator (`MANIFEST.md`), and the byte-level correctness cross-check independently confirms that all layers return identical responses before any timing is trusted.

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